

22PH202 - ELECTROMAGNETISM AND MODERN PHYSICS

UNIT III – LM 3.1- ELECTROMAGNETISM BASIC LAWS

1.1 INTRODUCTION

The operation of electric motors is governed by various laws of electricity and magnetism, including Faraday's law of induction, Ampère's circuital law, Lenz' law, and the Lorentz force. The first two – Faraday's law and Ampère's circuital law – are included in Maxwell's equations. Together with Lenz' law and the Lorentz force, these principles form the basis of electromagnetism.

1.2 FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

1.2.1 Faraday's law of electromagnetic induction: (referred to as **Faraday's law**) is a basic law of electromagnetism predicting how a magnetic field will interact with an electric circuit to produce an electromotive force (EMF). This phenomenon is known as electromagnetic induction..

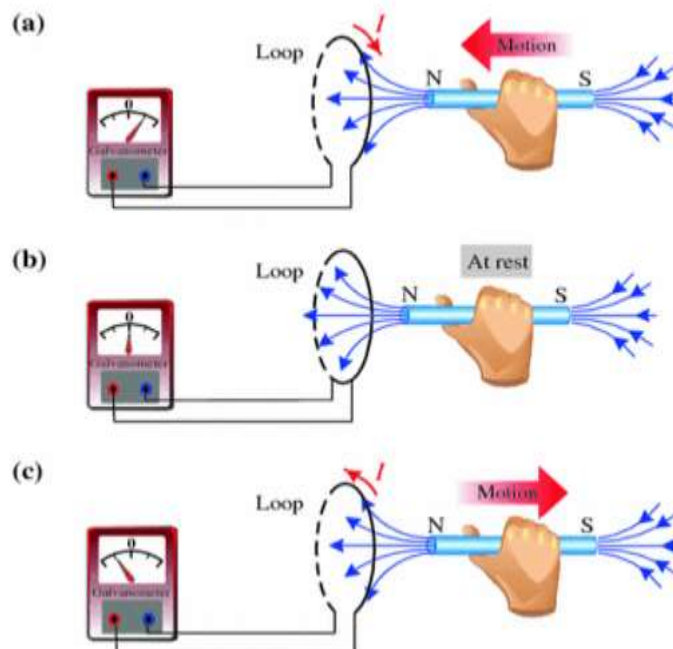


Figure-1 Faradays law of Induction

From the Figure 1, Faraday's law states that a current will be induced in a conductor which is exposed to a changing magnetic field. Lenz's law of electromagnetic induction states that the direction of this induced current will be such that the magnetic field created by the induced current **opposes** the initial changing magnetic field which produced it. The direction of this current flow can be determined using Fleming's right-hand rule.

Position of magnet	Deflection in galvanometer
Magnet at rest	No deflection in the galvanometer
Magnet moves towards the coil	Deflection in galvanometer in one direction
Magnet is held stationary at same position (near the coil)	No deflection in the galvanometer
Magnet moves away from the coil	Deflection in galvanometer but in the opposite direction
Magnet is held stationary at the same position (away from the coil)	No deflection in the galvanometer

1.2.2 Faraday's First Law Of Electromagnetic Induction :

Whenever a conductor is placed in a varying magnetic field, an emf is induced. If the conductor circuit is closed, a current is induced, known as the induced current.

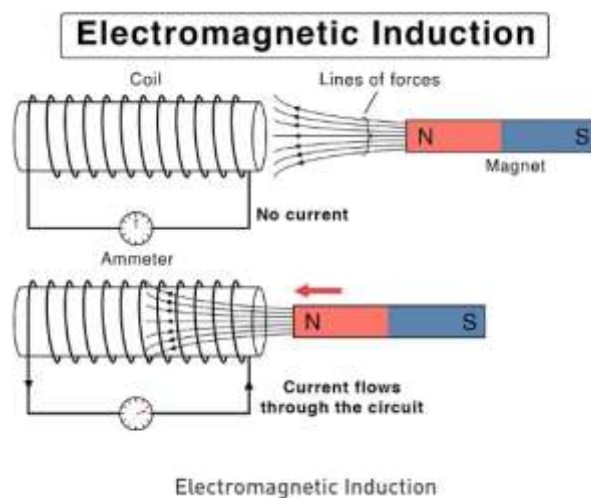


Figure-2 Electromagnetic Induction

1.2.3 Change of Magnetic Field Intensity inside a Closed Loop

In a closed loop, no magnetic field will be induced unless there is a power supply available. If a power supply is provided, the magnetic field will expand along with the current and will continue till the magnetic field collapses. Hence, magnetic field lines induced inside a closed loop are continuous.

Ways of Changing the Magnetic Field

There are specifically four ways to change the magnetic field inside a circuit:

- By modifying a region of a coil that is generally placed in the magnetic field.
- By moving the coil inside the magnetic field coil or outside the magnetic field.
- By moving the magnet in the same direction of a coil or against the coil's direction.
- By rotating the coil in relation to the magnet.

1.3 . Ampere's Law

Ampere's law, or Ampere's circuital law, is a mathematical statement used in electromagnetism that gives a relationship between a current and the magnetic field it generates. The magnetic field produced by an electric current is proportional to the magnitude of the current with a proportionality constant equal to the permeability of free space (μ_0), a universal constant in physics. Its value is $4\pi \times 10^{-7}$ H/m.

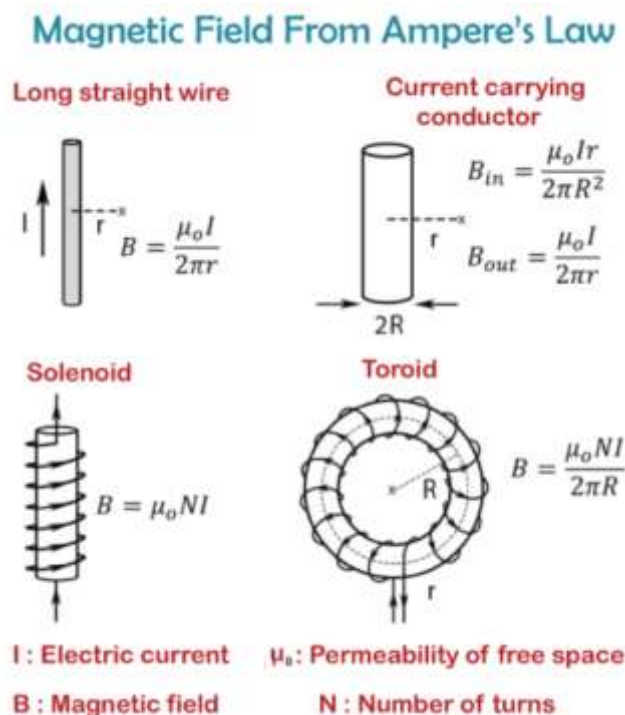


Figure-3 Magnetic field from Ampere law

From the Figure 3, According to Ampere's law, magnetic fields are related to the electric current produced in them. The law specifies the magnetic field that is associated with a given current or vice-versa, provided that the electric field doesn't change with time.

Ampere's Law can be stated as:

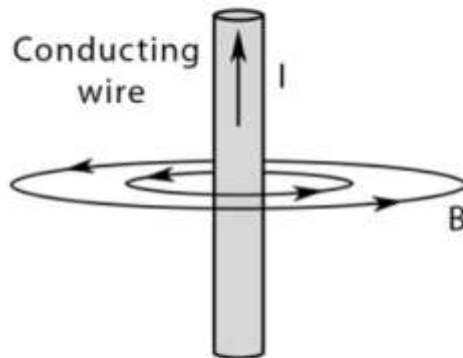
"The magnetic field created by an electric current is proportional to the size of that electric current with a constant of proportionality equal to the permeability of free space.

The equation explaining Ampere's law which is the final Maxwell's equation is given below:

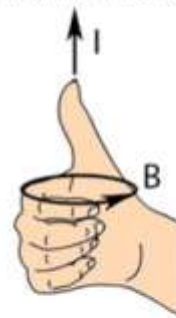
$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Maxwell's Equation

Ampere's Law



Right hand thumb rule



Integral form: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

Differential form: $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$

I : Electric current

B : Magnetic field

μ_0 : Permeability of free space

J : Current density

Thumb points in the direction of the electric current and fingers curl around the current indicating the direction of the magnetic field

1,3,1 Applications of Ampere's Law

Ampere's Law is used to :

- ☐ Determine the magnetic induction due to a long current-carrying wire.
- ☐ Determine the magnetic field inside a toroid.
- ☐ Determine the magnetic field created by a long current-carrying conducting cylinder.
- ☐ Determine the magnetic field inside the conductor.
- ☐ Find forces between currents

1.4 Lenz's Law

Electromagnetic induction is governed by two fundamental laws – Faraday's Law and Lenz's Law. Faraday's Law establishes a relationship between the induced emf (ε) and the magnetic flux rate ($d\phi/dt$) in a conducting coil of N turns. It is given by the following formula.

$$\varepsilon = - N d\phi/dt$$

Lenz's Law states,

“The polarity of the induced emf is such that it opposes the change in magnetic flux that produced it.”

When a magnetic field induces a current in a conducting coil, the induced current generates its magnetic field, opposite to the inducing magnetic field. In other words, an induced current will always oppose the motion that started it in the first place. Lenz's Law is significant since it can determine the direction of the induced current and the magnetic field induced by the current.

The change in the magnetic flux around a conducting coil may be caused in several ways:

- Change the magnetic field strength
- Move the magnet toward or away from the coil
- Move the coil into or out of the magnetic field
- Rotate the coil relative to the magnet

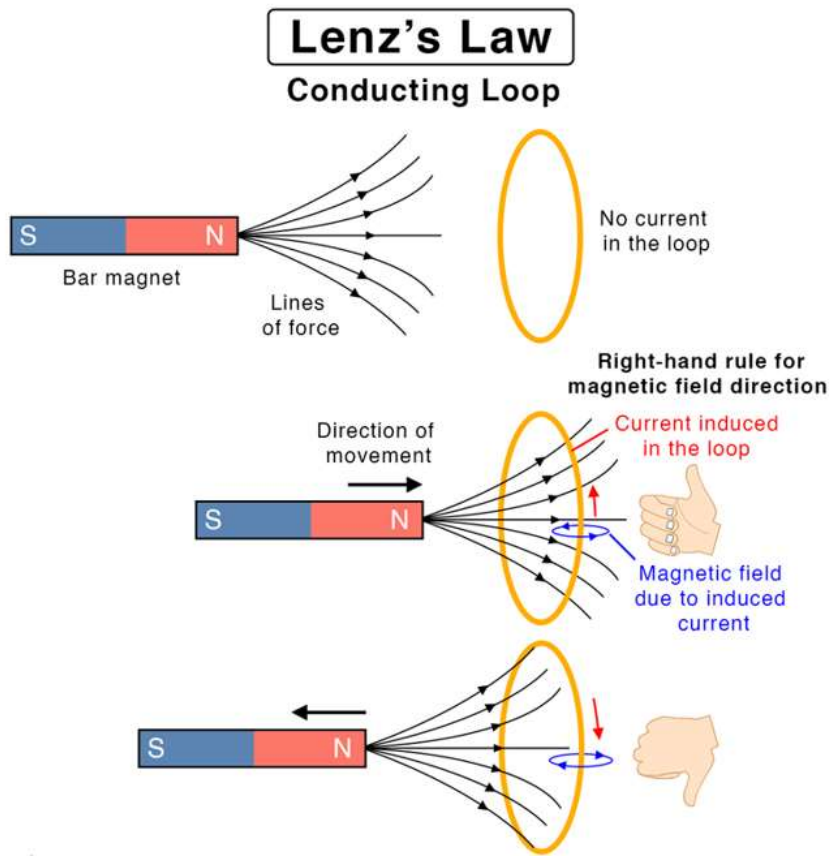


Figure-4 Lenz law

1.4. 1 Applications of Lenz's Law

Lenz's law can be applied to the following devices:

- Inductor
- Electric generators
- Electromagnetic braking
- Induction cooktop
- [Eddy current](#) equalizers
- Eddy current dynamometers
- Microphones
- Card readers
-

1.5 Introduction to Lorentz Force

Experts define Lorentz force as the combination of the magnetic and electric force. Furthermore, this force acts on a point charge due to electromagnetic field.

Lorentz force explains the equations of mathematical nature along with the physical importance of forces which act on the charged particles. Moreover, these particles travel through space which contain

Lorentz force refers to a combination of magnetic and electric force that acts on a point charge due to the presence of electromagnetic fields. Furthermore, the Lorentz force is also known by experts as the electromagnetic force.s electric and magnetic field.

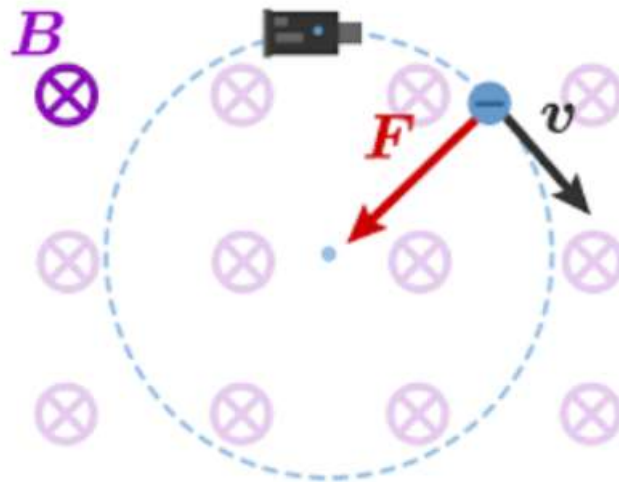


Figure-5 Lorentz force

Lorentz force on a moving charge that is present in a B Field

Lorentz force happens when the movement of a charged particle takes place through a [magnetic](#) field and cuts through field lines in the process. This force acts at right angles to both the particle velocity, v , and the magnetic field, B .

The formula of Lorentz Force

$$F=q(E+v*B)$$

Where,

- F is the force whose effect takes place on the particle
- q is the particle's electric charge
- v is the velocity
- E refers to the external electric field

- B is the magnetic field

1.5.1 Importance Of The Lorentz Force

Lorentz's force explains the mathematical equations along with the physical importance of forces acting on the charged particles that are travelling through the space containing electric as well as the magnetic field. This is the importance of the Lorentz force.

1.5.2 Right-Hand Rule

The right-hand rule is useful to find the magnetic force as it becomes easy to visualize the direction as given in Lorentz force law.

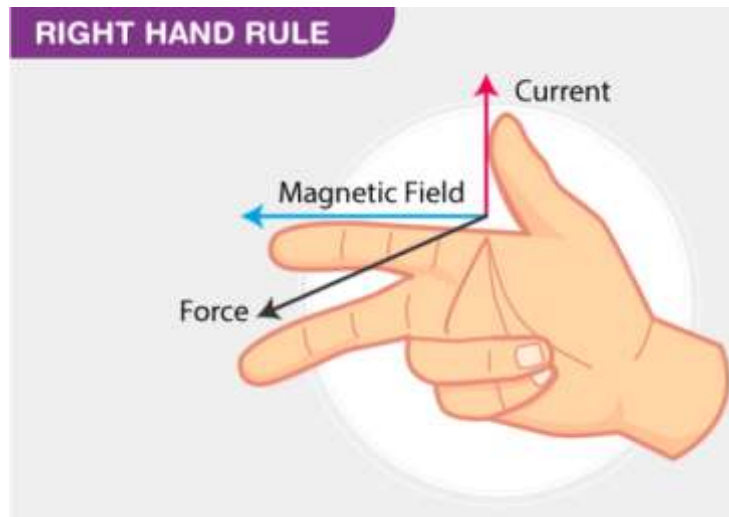


Figure-6 Right-Hand Rule

From the above figure, it is understood that the magnetic force is perpendicular to both the magnetic field and charge velocity.

1.5.3 Applications of Lorentz Force

The following are the applications of Lorentz force:

- Cyclotrons and other particle accelerators use Lorentz force.
- A bubble chamber uses Lorentz force to produce the graph for getting the trajectories of charged particles.
- Cathode ray tube televisions use the concept of Lorentz force to deviate the electrons in a straight line so they land on specific spots on the screen.